

GitHub Actions - CD

1. CD 개념

- CD(Continuous Deployment): 테스트를 통과한 코드를 자동으로 서버에 배포하는 것
- CI(테스트)가 성공한 뒤에만 이어서 실행 → "테스트 안 거친 코드가 배포되는" 사고 방지
- 흐름: 코드 push → **CI** 테스트 실행 → 성공 시 **CD** 배포 (Docker 이미지 빌드 → Docker Hub 푸시 → EC2 재배포)

2. 워크플로우 트리거: `workflow_run`

`.github/workflows/django-with-aws-deploy.yml`

```
on:
  workflow_run:
    workflows: ["Test (Django with aws)"]
    types: [completed]
    branches: ["main"]

jobs:
  deploy:
    if: github.event.workflow_run.conclusion == 'success'
```

- `workflow_run` : 다른 워크플로우(여기선 CI)가 끝났을 때 트리거됨
- `if: ... == 'success'` : CI가 성공했을 때만 배포 잡을 실행 (실패하면 배포 건너뛰)

3. Docker 이미지 빌드 & Docker Hub 푸시

```
- name: Docker Hub 로그인
  uses: docker/login-action@v3
  with:
    username: ${ secrets.DOCKERHUB_USERNAME }
    password: ${ secrets.DOCKERHUB_TOKEN }

- name: 이미지 빌드 & Docker Hub 푸시
  uses: docker/build-push-action@v5
  with:
    context: "Github Actions/Django with aws"
    push: true
    tags: ${ secrets.DOCKERHUB_USERNAME }/django-web:latest
```

- `docker build` → `docker login` → `docker push` 를 그대로 자동화
- 계정 정보는 코드에 직접 쓰지 않고 **Secrets**로 관리

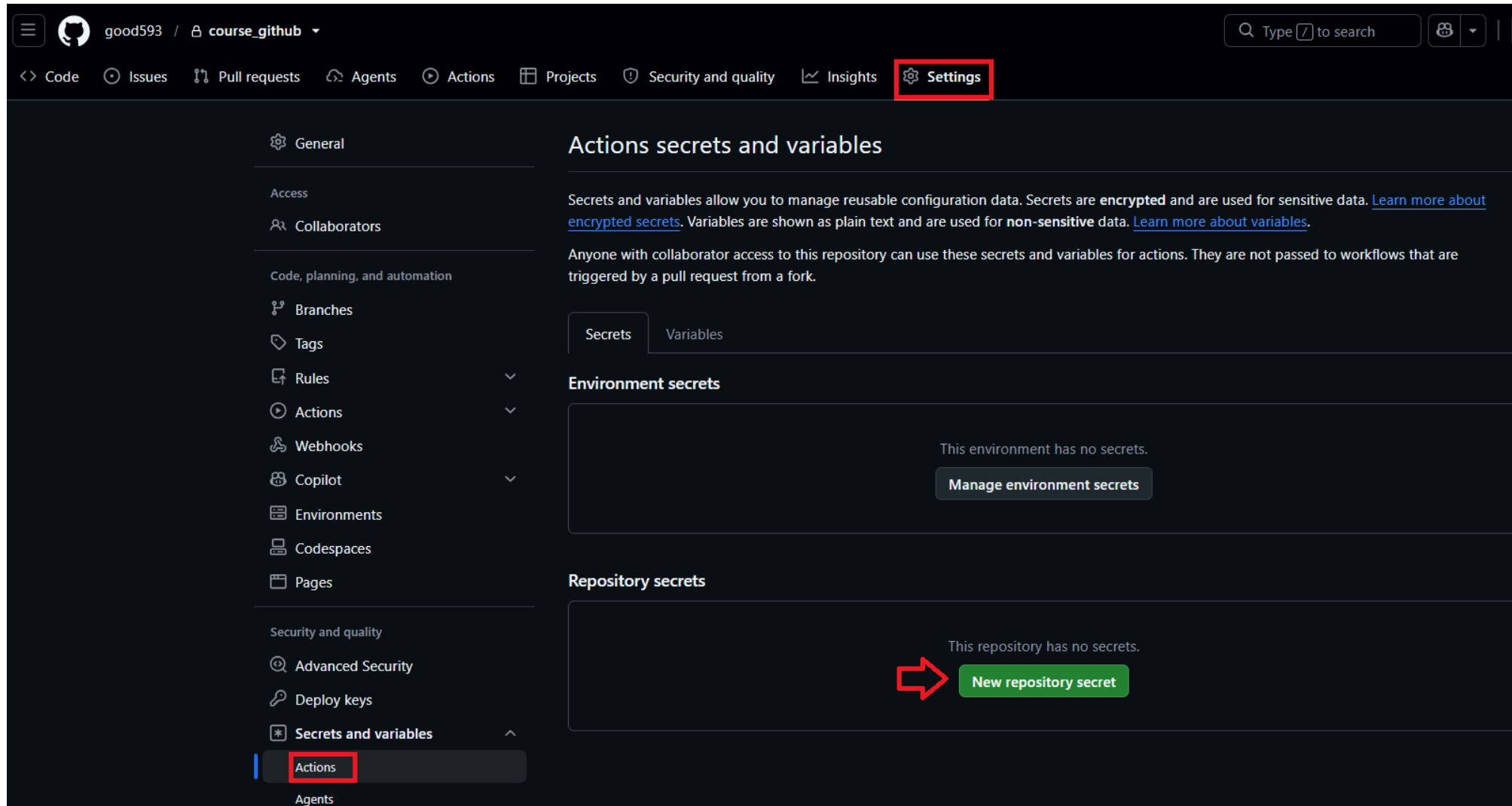
4. EC2에 재배포

```
- name: EC2 재배포
  uses: appleboy/ssh-action@v1
  with:
    host: ${ secrets.EC2_HOST }
    username: ${ secrets.EC2_USERNAME }
    key: ${ secrets.EC2_SSH_KEY }
    script: |
      docker pull ${ secrets.DOCKERHUB_USERNAME }/django-web:latest
      docker stop django-server || true
      docker rm django-server || true
      docker run -d --name django-server -p 80:80 ${ secrets.DOCKERHUB_USERNAME }/django-web:latest
```

- `appleboy/ssh-action` 이 GitHub Actions 러너에서 EC2로 SSH 접속을 대신 처리
- `script` : EC2 안에서 실행할 명령 — 최신 이미지 pull → 기존 컨테이너 정리 → 새 컨테이너 실행

실습

Actions secrets and variables



The screenshot shows the GitHub interface for the repository 'course_github' by user 'good593'. The 'Settings' tab is selected in the top navigation bar. On the left sidebar, the 'Actions' section is highlighted. The main content area is titled 'Actions secrets and variables'. It contains a description of secrets and variables, a 'Secrets' tab (selected), and two sections: 'Environment secrets' and 'Repository secrets'. Both sections indicate that there are no secrets currently. A red arrow points to the 'New repository secret' button in the 'Repository secrets' section.

good593 / course_github

Code Issues Pull requests Agents Actions Projects Security and quality Insights Settings

General

Access

Collaborators

Code, planning, and automation

Branches

Tags

Rules

Actions

Webhooks

Copilot

Environments

Codespaces

Pages

Security and quality

Advanced Security

Deploy keys

Secrets and variables

Actions

Agents

Actions secrets and variables

Secrets and variables allow you to manage reusable configuration data. Secrets are **encrypted** and are used for sensitive data. [Learn more about encrypted secrets](#). Variables are shown as plain text and are used for **non-sensitive** data. [Learn more about variables](#).

Anyone with collaborator access to this repository can use these secrets and variables for actions. They are not passed to workflows that are triggered by a pull request from a fork.

Secrets Variables

Environment secrets

This environment has no secrets.

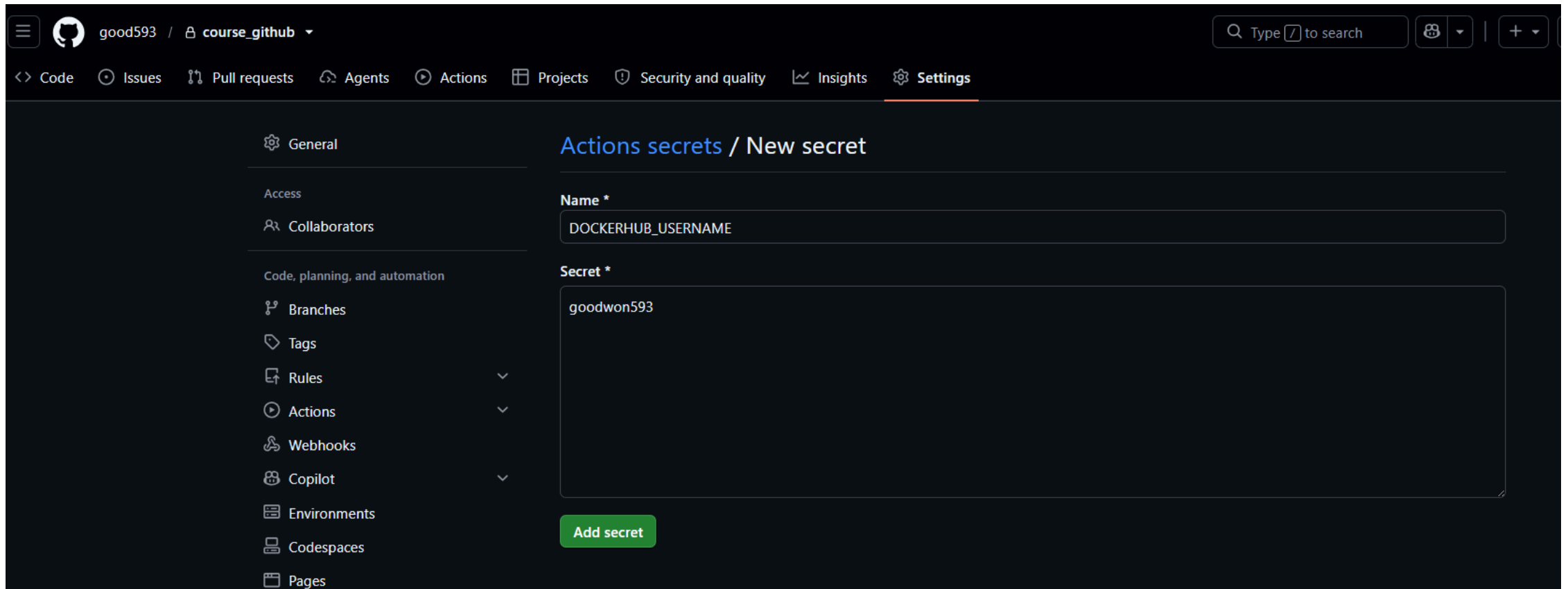
Manage environment secrets

Repository secrets

This repository has no secrets.

New repository secret

DOCKERHUB_USERNAME



The screenshot shows the GitHub Actions secrets page for the repository 'course_github' by user 'good593'. The 'Settings' tab is selected in the top navigation bar. On the left sidebar, the 'General' section is expanded, showing options like Access, Collaborators, and Code, planning, and automation. The 'Actions secrets / New secret' page is displayed, featuring a form with two fields: 'Name *' and 'Secret *'. The 'Name' field contains 'DOCKERHUB_USERNAME' and the 'Secret' field contains 'goodwon593'. A green 'Add secret' button is located at the bottom of the form.

good593 / course_github

Search Type to search

Code Issues Pull requests Agents Actions Projects Security and quality Insights Settings

General

Access

Collaborators

Code, planning, and automation

Branches

Tags

Rules

Actions

Webhooks

Copilot

Environments

Codespaces

Pages

Actions secrets / New secret

Name *

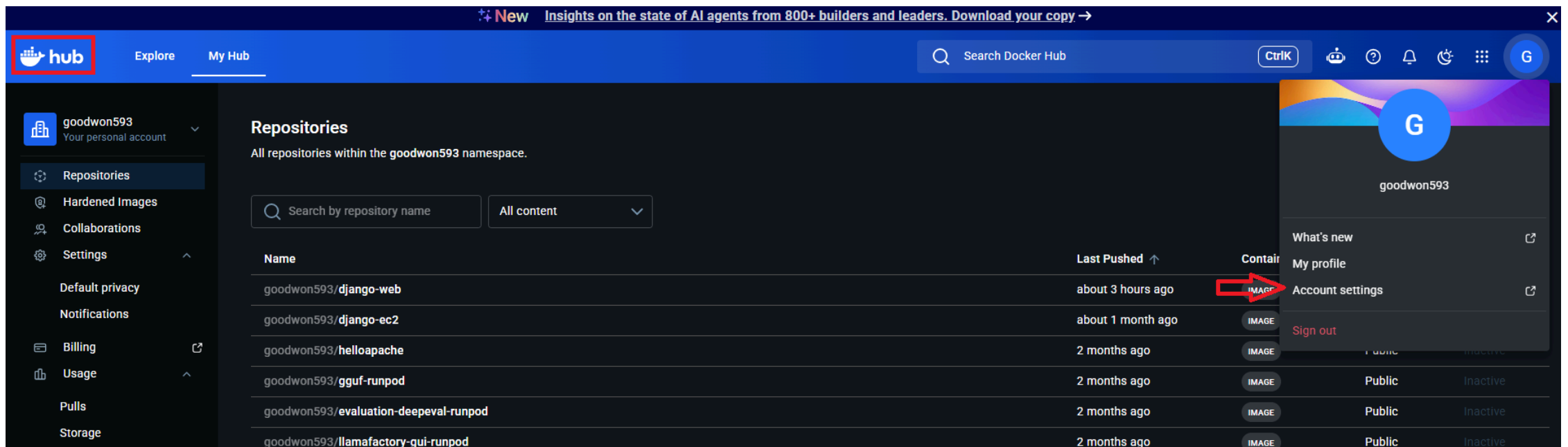
DOCKERHUB_USERNAME

Secret *

goodwon593

Add secret

DOCKERHUB_TOKEN



The screenshot shows the Docker Hub interface for a user named 'goodwon593'. The user's profile menu is open, displaying options: 'What's new', 'My profile', 'Account settings', and 'Sign out'. A red arrow points to the 'Account settings' option. The main content area shows a list of repositories under the 'goodwon593' namespace. The table includes columns for 'Name', 'Last Pushed', 'Container', 'Public', and 'Inactive'.

Name	Last Pushed	Container	Public	Inactive
goodwon593/django-web	about 3 hours ago	IMAGE		
goodwon593/django-ec2	about 1 month ago	IMAGE		
goodwon593/helloapache	2 months ago	IMAGE		
goodwon593/gguf-runpod	2 months ago	IMAGE	Public	Inactive
goodwon593/evaluation-deepeval-runpod	2 months ago	IMAGE	Public	Inactive
goodwon593/llamafactory-gui-runpod	2 months ago	IMAGE	Public	Inactive

New

Insights on the state of AI agents from 800+ builders and leaders. Download your copy →

docker

goodwon593

Your personal account

Home

Build Cloud

Account information

Email

Password

2FA

Personal access tokens

Connected accounts

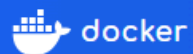
Convert

Personal access tokens

You can use a personal access token instead of a password for Docker CLI authentication. Create multiple tokens, control their scope, and delete tokens at any time. [Learn more](#)

Generate new token

Description	Scope	Status	Source	Created	Last used	Expiration date	
Generated through desktop a...	Read, Write, Delete	Active	Auto-generated	Jul 19, 2026 at 14:02:54	Jul 19, 2026 at 14:13:27	Never	
Generated through desktop a...	Read, Write, Delete	Active	Auto-generated	Mar 30, 2026 at 18:17:52	Jun 26, 2026 at 19:04:10	Never	
Generated through desktop a...	Read, Write, Delete	Active	Auto-generated	Apr 05, 2026 at 22:10:16	Apr 05, 2026 at 22:27:22	Never	



goodwon593

Your personal account

Home

Build Cloud

Account information

Email

Password

2FA

Personal access tokens

Connected accounts

Convert

Privacy

Deactivate

[Personal access tokens](#) / New access token

Create access token

A personal access token is similar to a password except you can have many tokens and revoke access to each one at any time. [Learn more](#)

Access token description

github-actions

Expiration date

30 days

Optional

Access permissions

Read & Write

Read & Write tokens allow you to push images to any repository managed by your account.

Cancel

Generate





goodwon593

Your personal account



Home



Build Cloud



Account information



Email



Password



2FA



Personal access tokens



Connected accounts



Convert



Privacy



Deactivate



Billing

[Personal access tokens](#) / New access token

Copy access token

Use this token as a password when you sign in from the Docker CLI client. [Learn more](#)

Make sure you copy your personal access token now. Your personal access token is only displayed once. It isn't stored and can't be retrieved later.

Access token description

github-actions

Expires on

Aug 18, 2026 at 23:59:59

Access permissions

Read & Write

To use the access token from your Docker CLI client:

1. Run

```
$ docker login -u goodwon593
```

Copy

2. At the password prompt, enter the personal access token.

[Redacted password field]

Copy



[Back to access tokens](#)

good593 / course_github

Type / to search

<> Code

Issues

Pull requests

Agents

Actions

Projects

Security and quality

Insights

Settings

General

Access

Collaborators

Code, planning, and automation

Branches

Tags

Rules

Actions

Webhooks

Copilot

Environments

Codespaces

Pages

Actions secrets / New secret

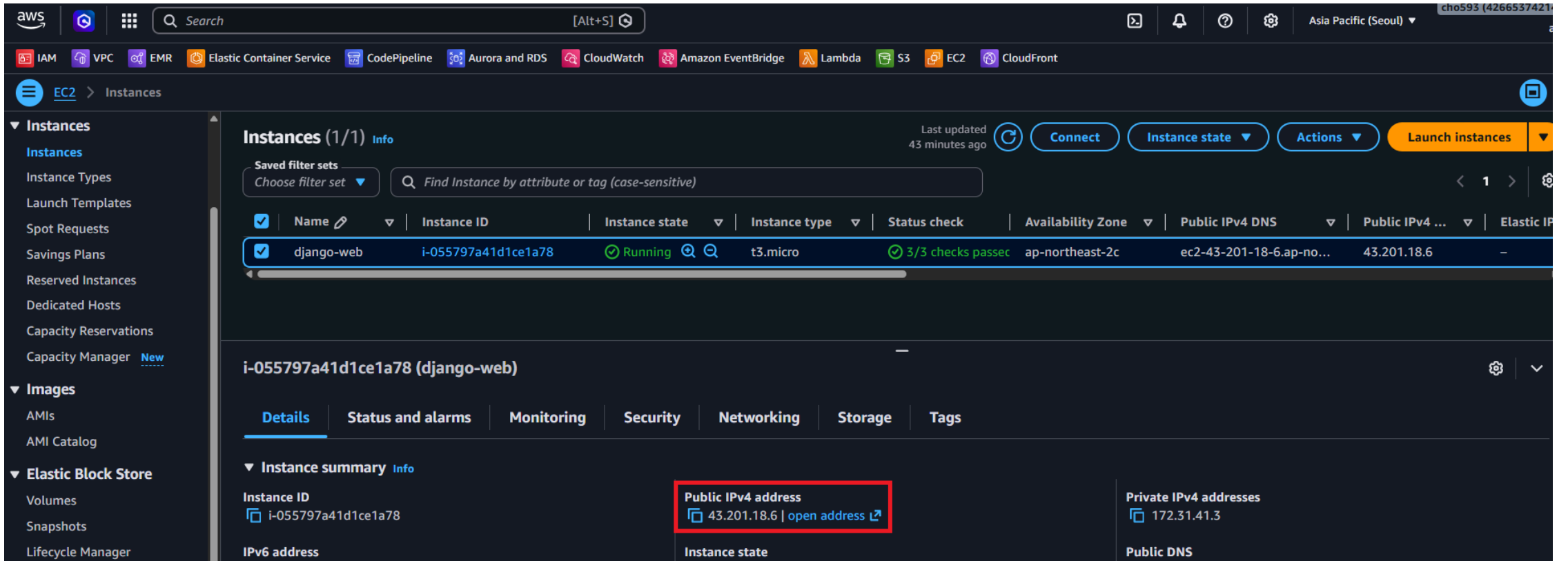
Name *

DOCKERHUB_TOKEN

Secret *

Add secret

EC2_HOST



The screenshot displays the AWS Management Console interface for the EC2 service. The left sidebar shows the navigation menu with categories like Instances, Images, and Elastic Block Store. The main content area shows the 'Instances (1/1)' page, which includes a table of instances and a detailed view for the selected instance 'i-055797a41d1ce1a78 (django-web)'. The instance is in the 'Running' state, and its public IPv4 address, 43.201.18.6, is highlighted with a red box. The console also shows the instance's details, including its ID, public DNS name, and private IP address.

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
django-web	i-055797a41d1ce1a78	Running	t3.micro	3/3 checks passed	ap-northeast-2c	ec2-43-201-18-6.ap-no...	43.201.18.6	-

i-055797a41d1ce1a78 (django-web)

Instance summary

Instance ID	Public IPv4 address	Private IPv4 addresses
i-055797a41d1ce1a78	43.201.18.6 open address	172.31.41.3

IPv6 address | **Instance state** | **Public DNS**

 good593 /  course_github

Type  to search

 | 

<> Code

⌚ Issues

 Pull requests

 Agents

⌚ Actions

 Projects

 Security and quality

 Insights

 Settings

 General

Access

 Collaborators

Code, planning, and automation

 Branches

 Tags

 Rules

 Actions

 Webhooks

 Copilot

 Environments

 Codespaces

Actions secrets / New secret

Name *

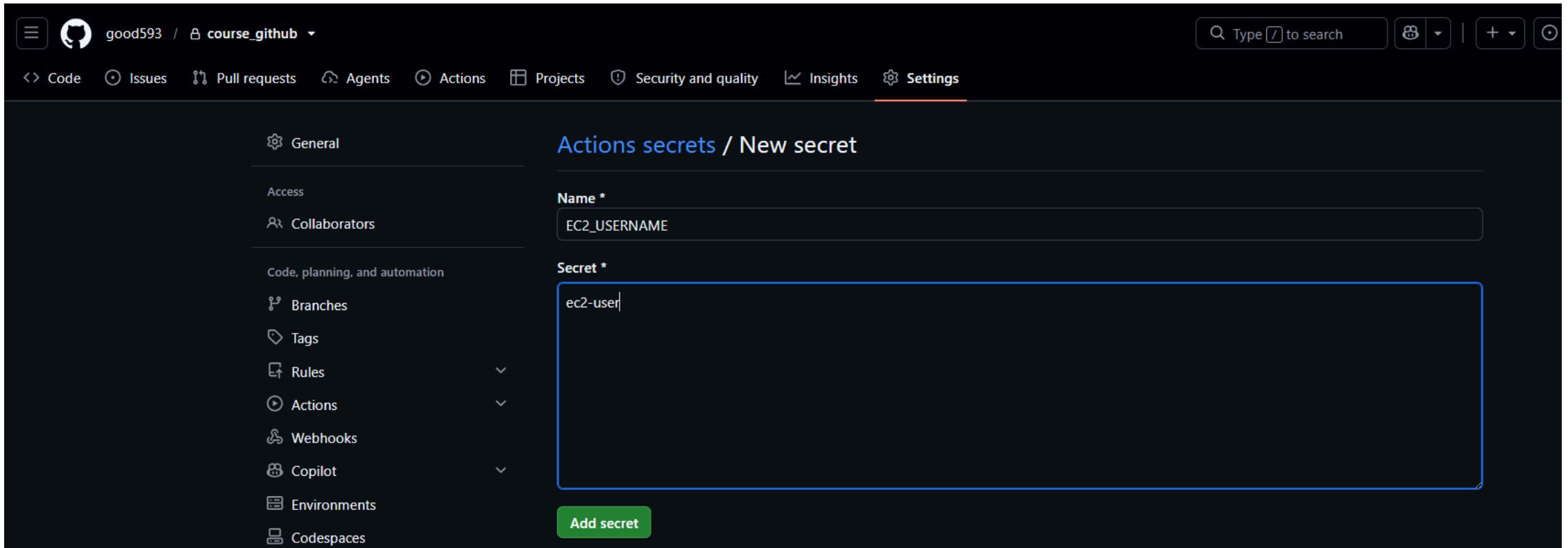
EC2_HOST

Secret *

43.201.18.6

Add secret

EC2_USERNAME



The screenshot shows the GitHub Actions secrets page for the repository 'course_github' by user 'good593'. The 'Settings' tab is selected in the top navigation bar. On the left sidebar, the 'Code, planning, and automation' section is expanded, showing options like Branches, Tags, Rules, Actions, Webhooks, Copilot, Environments, and Codespaces. The main content area is titled 'Actions secrets / New secret'. It contains two input fields: 'Name *' with the value 'EC2_USERNAME' and 'Secret *' with the value 'ec2-user'. A green 'Add secret' button is located at the bottom right of the form.

good593 / course_github

Code Issues Pull requests Agents Actions Projects Security and quality Insights Settings

General

Access

Collaborators

Code, planning, and automation

Branches

Tags

Rules

Actions

Webhooks

Copilot

Environments

Codespaces

Actions secrets / New secret

Name *

EC2_USERNAME

Secret *

ec2-user

Add secret

EC2_SSH_KEY

The image shows a code editor window on the left and a file explorer window on the right. The code editor displays the contents of a file named `django-keypair.pem`, which is a private key in PEM format. The file explorer shows the `Downloads` folder, with a red box highlighting the file `django-keypair.pem` and a red arrow pointing to it from the left.

```
-----BEGIN RSA PRIVATE KEY-----
MIIEowIBAAKCAQEAsIvjhaGPdXVe0+Es1v/JUaeZvFYG2HFjVOKdyaAcU0cwy1hI
2//hZ03gyjbYZuJcsaRDkWNppGSJ6XMtihg/dXyik/XJQnRj1e3ZnQsg+WkMagWs
zWY13sDNhS7RjewBPnVTq+wPEHS524BMGIJuBhtm1phRPtYknoQ0S3WuVLGAohfq
fp2dgYWpLv98jnh1P40GmLseCj0SgQR8c6anJizkiIl1fTdQK4G0P3zRVNW4HPJt
NTLiTWqAWS1/VABEEAWPjbFM6khhts+1DuldwijiklydDb42uCTRxbp/kR0z+HHx
MYt/9jEamfT4dBEQgwX70mAnY92zIhCBJ7ER9wIDAQABAoIBAIEIK0n32xfemhGa2
YV08M8875yKw+7aUFBh4/ZWjM3SAR5yJ1heWTPa4ldKPWbyDCTHaxPvq0GaqxXxy
/aIfF/4XNaor07HsYDxy0WkPhjhGt8IhhVcA2T402dLshK+WjZCTy00Q7UG2qKAY
Cl9Qy1+1eG7mY94htWkna3kG+MPtUwFw+gFfSB1DG0DiM763kgku/dqvGpdgbzTL
ezg4+0QXlhWrAAFChCoFLP5M33sFLgxbiay/5gqQT0ipsWaKpZ6BoZKPP0UP3VX/
mP3oI1qJ+2ZBky+VbdXred7i/dXnFKZEIA3XA1qex0JvtbcpAje1YUigG+30Igpz
/EoYBQECgYEA6KbvpCsXJpCd/1bDRx5p72whbH9GQxYn0yGAfIHkYBvoG0tcc0fV
RJoeW6Mh0E7FFohudaIx8kEs+M7q77asPr+EdJIGjWxFmRtYRDBMRPEH2LtdPm0c
xIKZjf0tvXzWJ0SqPFLYNPCV10Rx1bepx2Spspro7hDC0eXyf7xNNNcCgYEAwK0K
5XC1RZCYntBeE5h0Q2fckP0apIDFG7tLh0cTlBW05xcnr8EoHYA+pEv5txklmdrF
pd8sWRH630ky9bHWC3vzl6kT3FScfPHgSm095J6syr7tumw/E0/RhS4K5FH11YGK
PZMVP7oxHlmsfCymLMANFRGGdrP32FC1br+CR+ECgYEAzH9gCYeu08Euf/C9uVes
```

Downloads

django-keypair.pem

admin_credentials.csv

good593 / course_github

🔍 Type / to search

+

<> Code

🕒 Issues

🔗 Pull requests

👤 Agents

🔄 Actions

📁 Projects

🛡 Security and quality

📊 Insights

⚙ Settings

⚙ General

Access

👤 Collaborators

Code, planning, and automation

🔗 Branches

🏷 Tags

📄 Rules

🔄 Actions

🔗 Webhooks

👤 Copilot

📁 Environments

💻 Codespaces

Actions secrets / New secret

Name *

EC2_SSH_KEY

Secret *

-----BEGIN RSA PRIVATE KEY-----
eJ0rq/i7w+BHpTe/frGiiVc8hDt9I7hkWI+4cGkQ6BJcSL/JPsqst+w+IX2P2KFT
CdZCCs0IYWVhqND2ZwmllLxUz1byArtbvAen/bzbZcHDYa51TJKkC/SIXvpmbRsB
azi6SToNIKUudEIBQ/I6UwUCgYBHmxfunM3kl3jT85hJxs9MLVNksO6WpnSBVVdw
xM09r3ieS9WwHhXIUGZFD1A7ooRBkvfJvWqeZlc/yL6E/Ak28HiltaUBeRq5n3qa
VlwU4jw0+42rQs04U/26Bo+zO1bPXVF0guu5Mst6MXm3x8I1A1GPK4eR56J61Lc2
DFdcYQKBgBWc8eYroWjAYxyil4sZpzRO0T+M/mdMTGdhnsVEG95XLUtH6pojDA
148FSzt4sFkimnWNwNsTKi7Cjblef7nXUupuFaqtEseRUKTVucExdbO6J+8il09b
Ww3HhWWwgKULOTD8eBPPDBYJdz+vX/IXqM5ZS/p9gyBw/J670gp2
-----END RSA PRIVATE KEY-----

Add secret

GitHub Secrets 확인

Secret	용도
DOCKERHUB_USERNAME	Docker Hub 계정명
DOCKERHUB_TOKEN	Docker Hub 액세스 토큰 (비밀번호 대신 사용)
EC2_HOST	EC2 인스턴스의 퍼블릭 IP / 도메인
EC2_USERNAME	EC2 접속 계정 (예: ubuntu)
EC2_SSH_KEY	EC2 접속용 SSH 개인키(.pem 내용)

Branches

Tags

Rules

Actions

Webhooks

Copilot

Environments

Codespaces

Pages

Security and quality

Advanced Security

Deploy keys

Secrets and variables

Actions

Agents

Codespaces

Dependabot

Integrations

SecretsVariables

Environment secrets

This environment has no secrets.

Manage environment secrets

Repository secrets

New repository secret

Name	Last updated
DOCKERHUB_TOKEN	7 minutes ago
DOCKERHUB_USERNAME	14 minutes ago
EC2_HOST	4 minutes ago
EC2_SSH_KEY	1 minute ago
EC2_USERNAME	4 minutes ago

CD 실행

main 브랜치 커밋하면 자동 실행

The screenshot shows the GitHub Actions interface for the repository 'course_github' by user 'good593'. The top navigation bar includes links for Code, Issues, Pull requests, Agents, Actions, Projects, Security and quality, Insights, and Settings. The 'Actions' tab is selected, showing a list of workflows on the left sidebar: 'All workflows', 'Deploy (Django with aws)', 'Test (Django with aws)', 'Management', 'Caches', 'Attestations', 'Runners', 'Usage metrics', and 'Performance metrics'. The main area displays 'All workflows' with a search bar and a table of workflow runs. The table shows 4 workflow runs. The first run is 'Deploy (Django with aws)' which is completed. The second run is '작성중' (In Progress) for 'Test (Django with aws)' on the 'main' branch, committed by 'good593'.

Workflow	Event	Status	Branch	Actor
Deploy (Django with aws)		Completed		good593
작성중		In Progress	main	good593

good593 / course_github

Type to search

<> Code

Issues

Pull requests

Agents

Actions

Projects

Security and quality

Insights

Settings

Deploy (Django with aws)

Deploy (Django with aws) #1

Re-run all jobs

Summary

All jobs

deploy

Run details

Usage

Workflow file

Triggered via workflow run 3 minutes ago

Status

good593 completed 9c91dcd

Success

Total duration

1m 51s

Artifacts

—

django-with-aws-deploy.yml

on: workflow_run

deploy

1m 49s

Annotations

1 warning

← Deploy (Django with aws)

✓ Deploy (Django with aws) #1

Re-run all jobs

Summary

All jobs

deploy

Run details

Usage

Workflow file

> Annotations

1 warning

deploy

succeeded 1 minute ago in 1m 49s

Search logs

> ✓ Set up job

> ✓ 저장소 코드 체크아웃

> ✓ Docker Hub 로그인

> ✓ 이미지 빌드 & Docker Hub 푸시

> ✓ EC2 재배포

> ✓ Post 이미지 빌드 & Docker Hub 푸시

> ✓ Post Docker Hub 로그인

> ✓ Post 저장소 코드 체크아웃

> ✓ Complete job